

Web Development Notes (Simple + Clear + Definitions)

1. Semantic Tags

Definition: Semantic tags are HTML tags that clearly describe the meaning of the content inside them.

- 1 header = top part of page
- 2 footer = bottom part
- 3 section = group of content
- 4 article = independent content

Why use: Improves readability, SEO, and structure.

2. Block Elements

Definition: Block elements take full width of the screen and always start on a new line.

- 1 Examples: div, p, h1, section, article
- 2 Used to create layout structure

3. Meta Tags

Definition: Meta tags provide hidden information about the webpage to browsers and search engines.

- 1 Not visible on webpage
- 2 Used by Google to show search results
- 3 Example: description of website

4. Reflow and Repaint

Reflow Definition: When layout changes (size, position), browser recalculates structure.

Repaint Definition: When only style changes (color), browser updates appearance.

- 1 Reflow = heavy operation
- 2 Repaint = lighter operation

5. Storage Types

localStorage: Stores data permanently until manually cleared.

sessionStorage: Stores data temporarily and clears when tab closes.

Cookies: Small data stored and sent to server with every request.

- 1 localStorage use: theme, preferences
- 2 sessionStorage use: login session
- 3 cookies use: authentication

6. JWT Tokens

Definition: JSON Web Token is a secure token used to identify a user after login.

- 1 Access Token: short expiry, used for API requests
- 2 Refresh Token: long expiry, used to get new access token

7. Z-Index

Definition: z-index controls the vertical stacking order of elements (which appears on top).

- 1 Higher value = on top
- 2 Works only with positioned elements

8. Flexbox vs Grid

Flexbox Definition: A 1D layout system used to arrange items in a row or column.

Grid Definition: A 2D layout system used to arrange items in rows and columns.

- 1 Flexbox: best for small layouts
- 2 Grid: best for full page layouts